

Lec 54 Foundations of Natural Language Processing

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Transcript

[00:00] Hello everyone, welcome back to the lecture series on fundamentals of generative AI and large language models. In my today's session, we will discuss about the foundations of natural language processing. Okay. So the contents for today's session goes like introduction to NLP, application of NLP, NLP pipeline, text pre-processing, tokenization, stop word removal, limitization, text representations which includes glove to word to vec word embeddings and glove embeddings and then challenges with L N L N L N L N L N L N L N L N L N L N LP. Okay. So this is today's session is basically the foundations or just a brush up of the NLP found uh you know fundamentals because based on this one we will

[00:51] proceed to learn about the transformers the large language models and the advanced topics. Okay. So let us take today's session to have a quick look or a review of what are the fundamental processes that are involved in NLP. Okay. So what is NLP? Basically it is a technology that allows computers to interpret, manipulate and comprehend human language. So it is very well known that when a human speaks the computer needs to understand that one. it is difficult for the computer to understand human text or speech. So that is where NLP comes into picture and NLP you know

do does all the processing that is required for the human text to be changed into a form where the computers or the models computer you know deep learning models

[01:49] or architectures they understand. Okay. So if we have a human language text or speech input then it goes to NLP pipeline and we will see what is this pipeline later and then it gives some the model gives some meaningful response or action. So let us see what do I mean by meaningful responses or what do I mean by actions. So suppose my input is is this email spam. If the output is yes then the action is spam detection.

[02:25] If the input is the im the movie was amazing the output is a positive sentiment and the action it has taken here is sentiment analysis. If the input is hello and the output is hola it is machine translation. If the input is what is generative AI generative AI is a branch of AI that lands patterns from existing data so on and so forth. And the action is then question answering. Okay. So these are the various you know actions that can be or the applications that can be uh thought uh you know applications thought about in the terms of applications of NLP and these are the example inputs and these are the outputs that we expect from the model.

[03:09] Now a quick look. So NLP can be used for chat bots, translations, machine translations, question answering, summarization, spam detection, sentiment analysis. So these are few of the applications of NLP. There are a lot more applications and whenever a computer processes, understands or generates human language, NLP as is at work. Okay, human like language, then NLP is at task. Okay. So in NLP pipeline what we start with? We start with raw text. That text can come as a speech or a written thing as a as a written text. So that is a raw text. So for example, let us take I absolutely love this movie. Okay. So this is the raw text that you are having and then this is the input from the user. It can

[04:07] come as a form of text or a speech converted to text. Okay. But this is the raw text that we are having. Then it goes through something called as the text prep-processing. And in text prep-processing what we get? We get these three words absolutely loved movie. Now you may ask what happened to this you know I the then exclamation mark then the punctuations what happened to them. So we will see what exactly the text process pre-processing does in my next slides. Right now we will just keep it as text prep-processing and then we will see or we can just uh mention it here. It cleans and normalizes the text, removes unnecessary information and prepares text for analysis. So whatever unnecessary

[05:02] information was here in my raw text those have been removed in the text prep-processing and I have only the very meaningful words that have the information encoded in that you know those few words the information the entire information that is given by the raw text that is present in these three words absolutely loved movie okay and then the Next thing after text prep-processing the pre-processed text goes

to something called as the text representation. Okay. So we have to find out some representation. What is this representation? Basically as we all know that computers or models deep learning models or machine learning models they don't understand the text.

[05:56] So we have to feed the input in terms of some numeric numerical numbers and this text representation is nothing but the vector representation of each and every word. Okay. So this [clears throat] converts words into numerical vectors and enables computers to process the language. Okay. And then this goes for the you know your machine or the deep learning model and model does some action. Here I have taken sentiment analysis model and it learns the patterns from numerical representations and then what happens it prediction or output happens here. Okay. So what is the prediction or output that is happening here? It is a positive sentiment. Okay. So positive sentiment is coming out as the output of this

[06:46] model. Okay. So this is basically the NLP pipeline and we will see what exactly happens in you know text pre-processing in my uh next uh slide. So positive sentiment here it produces the final output and then we will see in the text pre-processing what is happening here. So as I was telling processing of or cleaning of the raw text into a structured format that can be easily analyzed by machines during your model performance. Okay, analyzed by machine learning and deep learning models. Okay, so this is that is what your text pre-processing does. So you have the raw text. Then what happens?

[07:34] The first thing that text prep-processing does does is lowering all the entire raw text and then it removes the punctuations and special characters because they don't give any you know vital information about the sentence. Okay. Then there is something called as tokenization which we will look into in my next slide. It is a very important part of text prep-processing. And then we do stop word removal. Basically removing the words which are not carrying information but they are if I keep them in my text which needs to be processed by my uh model there is a lot of computational cost. So I don't want my computational cost to increase without having any meaningful information coming from those words.

[08:33] Okay. So that is the process we call as stop word removal. We will see it in my next slide. And then there is another process called limitization which once again we will see. And then we get a clean text. Limitization basically changes all form of words to one standard form or baseline. Okay, clean text and then uh okay so we get the clean text. So here is an example. The raw text is the quick brown fox the the quick brown foxes are running. Okay. Visit <https://youknow.hashample.com>.

[09:17] Then there is a smiley. Then there are some you know starting uh you know punctuations here. So all these things are there. Now after pre-processing what we get? We get the quick brown fox run. Okay. So all the extra things that were here the

the smiley the punctuation marks the you know the link everything has gone. So that is what is the text prep-processing does. Okay. Now one of the most fundamental pre-processing step is tokenization where we break text into smaller units called tokens. Okay. So that is what we do in tokenization. As I was telling tokenization is a very very important part of a text prep-processing. So let's have a look into what is this tokenization. What exactly happens here.

[10:14] So it's a process of splitting text into smaller units called tokens. And these tokens can be words, sentences, subwords or even characters depending on the NLP application. Okay. So these are the things that can happen in tokenization. So let us have a look into what is this word tokenization. Suppose the example that I'm taking here is I love learning natural language processing. Now when I do word tokenization I am basically dividing each and every word into one one each token. So I love learning natural language processing.

[10:57] So this is called as tok word tokenization. What is sentence tokenization? Suppose I have a full paragraph. Here I have taken three um sentences for the example. NLP is fascinating. It enables computers to understand languages and it powers. Okay. So if I do sentence tokenization, I get three different sentences. NLP is fascinating. It enables computers to understand languages and it powers jativity. So this is called as sentence tokenization. That means each and every sentence is becoming a separate token.

[11:39] Then we have subword tokenization. What is that one? If we have a word which can be broken into subwords then we call it as subword tokenization. Like unbelievable. The word is unbelievable. Unbelievable. So these are my subword tokens. Okay. And these are used in modern modern LLMs such as birds and GPTs. We'll see when see them you know when we are studying about LLMs and BS and GPS. Okay. Then there is something called as the character tokenization. What is this character tokenization?

[12:20] Suppose I have the word NLP then the character token are N L and P. Okay. So breaking the word into separate separate characters as individual tokens. So that is character tokenization. Now tokenization why do we require tokenization? It converts raw text into smaller meaningful units or tokens so that it becomes easy for the model to understand. the information that is encoded in the text making it possible for computers to process human language effectively. Okay.

[13:00] Then we will look into what is this stop word removal. Basically stop words what are these stop words? They are the frequently used words which we see in any sentence like the is are, of and. If you see very carefully these words do not give us any any uh you know additional information but they are there. So if I remove these words what do I get? If I remove these words I get the meaning of the text there without increasing the computational cost of the model. That means if I have these

words is, are of and in my sentence then each and every word becomes a token and each and every word needs to be processed. There will be a text embedding. There will be a vector representation. Then the model will

[14:02] start you know doing all sorts of mathematical calculations without improving or without having any extra additional information. So those things can be treated as noise and can be removed because the main idea of the process is to understand what is there in the raw text. Okay. So if the original sentence is a students are learning natural language processing after stop word removal we get students learning natural language processing. So da is removed are is removed. So all these things have been removed and why it is required as I was telling stop words usually carry little semantic meaning and can be often removed to reduce noise in the text. So that is the reason why stop word removals are used

[14:55] in text prep-processing. Then we will see what is this limitization. If you if you remember the pipeline there was a block called as limitization. We will look into what exactly the limitization is. It is process of converting a word to its base or dictionary form. It is called as the lema while preserving its meaning. Okay. So all sorts of words can be converted into its base or dictionary form which we call as lema. We'll just see what is what do I mean by the original word and the lema.

[15:34] Suppose we have running. The running word comes from run. So the dictionary form is run. Studies study. If the original word is children, the lema is basically child. If the original one is better word is better, then the lema is good. Because better is just a superlative form of good. Good, better, best. But the actual base word is good. So we change from the original word to the lema using the process limitization.

[16:11] Now the question comes why it is required? It reduces different forms of the same word to a common base. Suppose I give better or best. It comes from the word good. So the semantic information is present in the word good. Suppose a person have five children or 10 10 children. The the basic word or the dictionary word is child. That means the person is having child. Okay, that number may be there. That is the reason the children came into picture. The plural form.

[16:50] So this is what happens in lematization and after pre-processing we have to see how the computer actually understand these words. So in the text prep-processing pipeline we have first understood what are the different text prep-processing steps and now after the text prep-processing is done we are looking into how okay now I have a text which is not having all the stop words I have removed them I have removed the punctuations I have changed any word to their base or the dictionary form and I have done all sorts of things here and then what it is still in the form of text. So what do I do so that my computer understands this text? Okay, that happens or the process by which I make my computer understand the text

[17:48] um you know information is called as text representation. So as I was telling

machines do not understand words or sentences. They have to be converted into some numbers before they are given to the model. Okay. And that is exactly happens that is exactly what happens in text representation. Now there are various traditional techniques of text representation like one hot encoding bag of words TF. We will not go into that those traditional techniques. What we will look into is the modern deep learning techniques which are used in the deep learning frameworks or generative AI frameworks.

[18:33] One of them is the word embeddings under which comes word to vec embedding and glove embedding and there is another one called as the contextual embedding under which falls B and GPT. Now this contextual embeddings like BERT and GPT we will take up when we are studying about the transformers we are clear about the transformer architectures for the current discussion we will restrict ourselves into the word embeddings that are produced by word tovec and glove embeddings. Okay.

[19:06] Now we will see what is this word tovec word embeddings. So what is the core idea? So basically here what happens whatever embedding it is it is taking the text and then it is changing into some numbers. Okay. And these numbers basically should have the semantic information that is present in your text. They should carry these are the footprint or the fingerprints. That semantic information should be present in the vectors or any numbers that we are feeding into the machine.

[19:42] Now the core idea is in word to vec learning happens from local context to windows and there are some vectors which are produced and which whose length and directions give the semantic information that is present in the text. So here is a image a very well-known image here and you know these are this is a x and yaxis this is the uh origin and we have certain vectors which represent the words king. So this vector is king this vector is for queen this vector is for man this vector is for woman.

[20:29] So we have certain vectors which starts from origin and they represent various words in the vocabulary. King, queen, man, woman. Okay. Now this red vector gives king minus man. What is your man? Man is here and this is your king. Basically king minus man plus if you do man then this is the you know this is the parallelogram law of vector addition king minus man plus man it gives you nothing the the resultant is basically your is your king vector. Okay. So this is your king minus man. This vector sorry.

[21:25] So this is your king minus man. This vector and this is your man. So if I add king minus man this one and your man here. So this is the parallelogram law of vector addition. You get you know this one then this one. So the resultant is king. Now the direction and length of the vectors encode the semantic properties of the word. Okay as I told. Now if you see this king minus man this is basically some male

royalty. Okay. So king is a male royalty. Okay.

[22:12] Now if you take this king minus man and you add woman to that one what will happen? So your king minus man. King is basically your male royalty royal figure. Okay. And [snorts] what happens is if I add woman to this king minus man vector. So this is my king minus man and this is my woman. So this parallelogram if you construct with king minus man and woman then the resultant is almost equivalent to your queen. That means king is a male royal figure.

[23:14] From that one if you take out king minus man and add woman there you it becomes queen. Okay. So which is the gender change is so that kind of information or that kind of semantic relationship is maintained in word to word embeddings. So we create the vectors in such a way the semantic relationship or semantic meaning of the word gets stored there in the form of vectors. Okay. So this is about what to word embeddings. Then we will look into glove embeddings. What are these glove embeddings? Basically the core idea is words that frequently occur together in a large corpus are likely to have similar meanings. So if there are two words like hot coffee, hot and coffee they come together frequently.

[24:14] That means there is some relationship there is some meaning relationship between hot and coffee. Okay. So there is some semantic relationship between them and learn dense word embeddings using word co-occurrence matrices. So what we do in glove is first we try to understand the frequency with which two words are coming together. When two words are coming together you know like cat and dog probably lot of times in animals we tell cat and dog if they are coming together that means there is some semantic information or semantic meaning which is making bringing them close and using that information we learn the dense word embeddings and these dense word embeddings are learned using something called as the

[25:10] co-occurrence matrix. So we look into what is this co-occurrence matrix. Suppose I have the words king, queen, man and woman. I'm trying to make a matrix which will tell me what is the you know how many times or how frequently my king and king are coming together. You know queen and king are coming together, man and king are coming together. woman and king are coming together or king and queen are coming together, king and man are coming together, king and woman and coming together so on and so forth. So if this is my co-occurrence matrix which gives king and king two kings you know side by side the number of times that I get in a large corpus is zero. King queen yes 120 times I have got king man 250 times king

[26:06] and a woman very less than queen king king 120 queen is zero queen man is 75 queen woman is 240 man king man and king coming together is 250 times man and queen coming together is less 75 man and man coming together is zero man and woman coming together is 110. Okay. So like this woman king is 80. Um you can see woman king was 80 here also. Same thing here. Woman and queen is 240. If

you see king and woman that number was here also. And then woman and man is 110 and man and you know um sorry and woman and woman is zero. Okay. So this is called as the co-occurrence matrix and using this co-occurrence matrix we try to understand what is there any semantic relationship or meaning which are bringing these words closer together

[27:11] and based on that one your word embeddings are produced. Okay. High co-occurrence values indicate stronger semantic relationship between words. Okay. So this is what is done in glove embedding. So now we will have a look we have seen how text representations are done but how text pre-processing text representations and then they are ready to go for the deep learning model for performing certain task. Now we will have a look into what are the challenges that are faced in NLP. One is the discrete nature of text data.

[27:52] Suppose you have two words sky and stars red and pink. So sky and stars they are having some relationship but that is not clearly you know doesn't tell anything about the similarity. Okay. Evolving nature of natural language is another thing. New words get added to the dictionary from time to time. Like example, rz is a new word which got added into the dictionary but we don't have any corpus that is having that word. Then ambiguity and vagueness.

[28:33] Never tasted a pizza like this before. What does it mean? It can be positive sentiment, can be negative sentiment. We don't know. There is an ambiguity. Also capturing long range dependency. This is a very important part like animal do not cross the road as it was tired or wide. Okay. So animals if you want to use wide then it should relate to the road. If you want to use tired that should be related to the animal. So this long range dependency capturing that one is was difficult in NLP and that is where our sequential learning of text comes into picture which is the topic of my next session.

[29:28] Okay. So with this one I come to the end of today's discussion. Before s before concluding let me summarize once what we have learned or what we have revised today in NLP basics. So we have seen what exactly is NLP what are the text prep-processing that we do to find out only the words that are required to keep the semantic information or the important information in a text. We remove all the unwanted things using stop word removal. bring them down uh to one base word using leatization. We remove all punctuations and other things and make it a clean text. And what is the reason for making a clean text?

[30:14] Because I want the clean text to be going forward for further mathematical manipulations using the deep learning models or mathematical processes in the deep learning model or any other model. so far uh you know uh any other model that comes into application and without increasing the computational cost. So that was the main idea for text prep-processing where we get a clean data. Then after text prep-

processing it has to go and get converted into some numerical numbers or vectors that is done using text representation. And we have seen two types of text representations like word tove vec and glove embeddings here. Okay, these are called embeddings. And these embeddings actually go into the deep learning

[31:06] models or your generative AI models because that is where all the math comes into picture. All math calculations come into picture and then finally we get some output. Okay. But one of the problems that was faced in NLP is capturing this long dependency and you know whether what is the way we can capture long range dependency and that is where our sequential processing of data was required using RNN and LSTMs having some memory of the past and that is what we will study in my next session.

[31:48] So my next session will be sequential modeling with RNN and LSTMs. Till that time please keep learning, happy learning. Thank you all.